Day 02: Networking Fundamentals

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Cybersecurity Learning Log

Part 1: Theory Questions and Answers

1. What is a computer network?

A computer network is a group of connected devices that communicate and share information or resources.

2. What is the difference between a client and a server?

A client requests a service or resource, while a server provides it. For example, a web browser requests a webpage from a web server.

3. What is a LAN?

A Local Area Network connects devices within a limited area, such as a home, office, or school.

4. What is a WAN?

A Wide Area Network connects networks over large geographical areas. The internet is a common example.

5. What is the purpose of a switch?

A switch connects devices on a local network and forwards Ethernet frames to the appropriate device using MAC addresses.

6. What is the purpose of a router?

A router forwards IP packets between different networks. A home router commonly connects a local network to the internet.

7. What is a firewall?

A firewall monitors and filters network traffic according to security rules, helping block unauthorized connections.

8. What is an IP address?

An Internet Protocol address identifies a network interface for communication using IP. IPv4 addresses are 32 bits long; IPv6 addresses are 128 bits long.

9. What is a MAC address?

A MAC address is a link-layer address used to identify a network interface on a local network. Its format and behavior depend on the network technology and configuration.

10. What is a default gateway?

A default gateway is the router or next-hop device a computer uses to send traffic when it has no more specific route for the destination.

11. What is DNS?

The Domain Name System translates domain names, such as example.com, into IP addresses and provides other information about domain names.

12. Why is networking knowledge important for cybersecurity?

Networking knowledge helps security analysts understand traffic, investigate suspicious connections, interpret IP addresses and ports, analyze DNS activity, identify exposed services, and troubleshoot security incidents.

Part 2: Practical Lab (Kali Linux)

The following screenshots were taken on a Kali Linux machine. They show the IP configuration of the machine, a connectivity test to a target host, and the path packets take to reach it.

Figure 1: Network Interface Configuration (ifconfig)

```
┌──(kali-linux㉿Kali-Linux)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:febf:2962  prefixlen 64  scopeid 0x20<link>
        inet6 fd17:625c:f037:2:9733:6ad7:8265:d4fd  prefixlen 64  scopeid 0x0<global>
        inet6 fd17:625c:f037:2:a00:27ff:febf:2962  prefixlen 64  scopeid 0x0<global>
        ether 08:00:27:bf:29:62  txqueuelen 1000  (Ethernet)
        RX packets 13  bytes 4131 (4.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 34  bytes 5292 (5.1 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

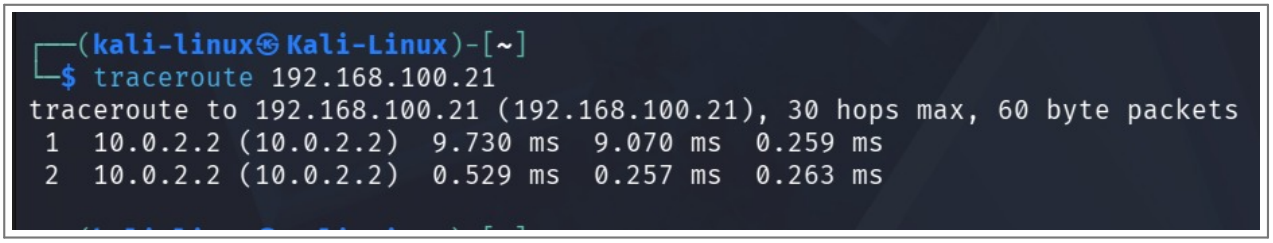
The **ifconfig** command shows the interface **eth0** is UP and RUNNING with the IPv4 address **10.0.2.15**, netmask 255.255.255.0 and broadcast 10.0.2.255. The MAC address is 08:00:27:bf:29:62. IPv6 addresses are also assigned, and lo is the loopback interface (127.0.0.1).

Figure 2: Connectivity Test (ping)

```
┌──(kali-linux㉿Kali-Linux)-[~]
└─$ ping 192.168.100.21
PING 192.168.100.21 (192.168.100.21) 56(84) bytes of data.
64 bytes from 192.168.100.21: icmp_seq=1 ttl=64 time=0.990 ms
64 bytes from 192.168.100.21: icmp_seq=2 ttl=64 time=0.785 ms
64 bytes from 192.168.100.21: icmp_seq=3 ttl=64 time=1.75 ms
64 bytes from 192.168.100.21: icmp_seq=4 ttl=64 time=0.981 ms
64 bytes from 192.168.100.21: icmp_seq=5 ttl=64 time=0.829 ms
64 bytes from 192.168.100.21: icmp_seq=6 ttl=64 time=0.822 ms
64 bytes from 192.168.100.21: icmp_seq=7 ttl=64 time=0.910 ms
64 bytes from 192.168.100.21: icmp_seq=8 ttl=64 time=0.977 ms
64 bytes from 192.168.100.21: icmp_seq=9 ttl=64 time=0.930 ms
64 bytes from 192.168.100.21: icmp_seq=10 ttl=64 time=0.765 ms
64 bytes from 192.168.100.21: icmp_seq=11 ttl=64 time=0.725 ms
^C
── 192.168.100.21 ping statistics ──
11 packets transmitted, 11 received, 0% packet loss, time 10014ms
rtt min/avg/max/mdev = 0.725/0.951/1.749/0.267 ms
```

ping 192.168.100.21 sent 11 packets and received 11 (0% packet loss). Reply time averaged about 0.951 ms (min 0.725 / max 1.749 ms) and TTL was 64, so the target host is reachable and responding.

Figure 3: Path to the Target (traceroute)



traceroute 192.168.100.21 shows packets leaving through **10.0.2.2**, which is the default gateway of this network, before reaching the destination.

Part 3: IP Address and DNS Server Summary

Item	Value	Source / Notes
Kali machine IP address	10.0.2.15	Figure 1 (ifconfig, eth0)
Netmask	255.255.255.0	Figure 1 (ifconfig, eth0)
MAC address	08:00:27:bf:29:62	Figure 1 (ifconfig, eth0)
Default gateway	10.0.2.2	Figure 3 (first hop in traceroute)
Target IP address	192.168.100.21	Figures 2 and 3 (ping and traceroute)
DNS server	Not visible in the screenshots	Check with **cat /etc/resolv.conf** (the nameserver line). Fill in here once confirmed.

IP address: identifies a network interface so that devices can send and receive packets. Here, Kali uses 10.0.2.15 and talks to the target 192.168.100.21 through the gateway 10.0.2.2.

DNS server: the server the machine asks to translate domain names (such as example.com) into IP addresses. Without a working DNS server, names fail to resolve even though IP connectivity works.